



VICTORIA UNIVERSITY OF
WELLINGTON
TE HERENGA WAKA

Constraining turbulence in the intracluster medium with the radio SZ effect

by

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CURRENT DEGREES



Candidate's ORCID

*A thesis submitted to the Victoria University of Wellington in partial
fulfillment of the requirements for the degree of Doctor of Philosophy*

Te Herenga Waka - Victoria University of Wellington

2027

Abstract

Gas motions. SZ effect Pressure fluctuations. Stable methodology. modeling. simulation and some interferometry.

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Acknowledgments

Praise to Allah that has given me strength. Appreciation to my parents for their support. Thanks to my supervisors for their patience.

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- Dedications (if applicable)
- Table of Contents
- List of Figures and Tables
- List of Abbreviations used in the thesis
- Main text of the thesis
- Bibliography or List of References
- Appendices

Date of thesis template release: 22 March 2019

For F.N.E. May she live forever.

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List of Abbreviations and Symbols

Abbreviations

AC	Alternating Current
AFM	Atomic Force Microscopy/Microscope
<i>etc.</i>	<i>etc.</i>

Symbols

$\hat{\rho}$	Density operator
<i>etc.</i>	<i>etc.</i>

Chapter 1

Scientific Landscape

“If you wish to create a Universe from scratch, the final ingredient would be clusters of galaxies.”

— [Afiq Hamid 2025]

1.1 Galaxy Clusters

It has been said that mass and time are among the most fundamental parameters in all of astrophysical science. This thesis is principally an exploration of the former through the study of the most massive virialized objects known to exist in the Universe: galaxy clusters.

Galaxy clusters occupy a unique position within the hierarchy of cosmic structures. They represent the endpoint of cosmic structure formation, populating the nodes that connect the filaments of the cosmic web [ref]. The composition of galaxy clusters is dominated by a hitherto enigmatic, collisionless dark matter component (80%), while the remaining baryonic component consists primarily of the intracluster medium, a diffuse and turbulent plasma (15%), and the luminous member galaxies that play host to stars, planets, and people (5%) [ref].

As massive virialized structures ($10^{14} - 10^{15} M_{\odot}$) the abundance of clusters can be used as probes for cosmology via the cluster mass function (CMF) which has the form:

$$\frac{dn}{dM} = f(\sigma_M) \frac{\bar{\rho}_m(z=0)}{M} \frac{d \ln \sigma_M^{-1}}{dM}$$

where $\bar{\rho}(z=0)$ is the present day mean matter density of the universe, σ_M characterizes the variance of density perturbations smoothed at a given mass scale M , and $f(\sigma)$ describes the non-linear collapse behaviour of dark matter halos. Considering parameters of the Λ CDM cosmology model, even small variations in cosmological parameters such as Ω_m or σ_8 may therefore induce measurable changes in the abundance of massive clusters [ref].

In practice empirical inference of the CMF is not a direct process but often carried out through a variety multi spectral traces of tracers that probe distinct physical components of the system. Hence the study of galaxy clusters relies on a multi spectral approach. *It is however, its not all sunshine and roses. Be warned that this thing is not measured directly. There are multiple empirical methods that attempt to constrain it each with its own advantages and limitations.*

1.2 Observables

Galaxy clusters appear differently based on the way they are observed. *The idea of clustering had been hinted at before prior to the works of George Abell but it was not until* CluseInitially seen in optical by George Abell which. Zwicky did some stuff too. Abell found the richness score, this helps comology how? Then revealed in X rays with satellites and then in radio SZ, this uncovered a wealth of findings such as the ICM. This work will focus on the SZ.

1.2.1 Radio Sunyaev Zeldovich effect

The radio SZ . Is a distortion in the

1.3 The cluster environment

1.4 Turbulence theory

1.5 Objectives moving forward

Hello I am here. The following publication has been incorporated as Chapter ??.

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If your task breakdown requires further clarification, do so here. Do not exceed a single page.

Chapter 2

Assumptions and Transfer Effects in SZ Surface Brightness Fluctuation Analysis

In this chapter I discuss the key principles of what goes into the turbulence inference pipeline.

2.1 Smooth SB model

The smooth SB model is a 2D picture of the cluster sans any turbulence. It is essentially a picture of cluster in the state of HE. Consider the Reynolds decomposition for turbulence.

$$S_{\text{obs}}(\theta) = \bar{S}(\theta) + \delta S(\theta)$$

Where $S_{\text{obs}}(\theta)$ is the observed surface brightness, $\bar{S}(\theta)$ is the smooth SB model, and $\delta S(\theta)$ is a small fluctuations term.

How has smooth models been defined before?

What is the problem of these models. We can do elliptical as well. What is problem with that.

2.1.1 Smooth SB model

2.1.2 Smooth SB model

Hello I am here. The following publication has been incorporated as Chapter 3.

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Chapter 3

Methodologies for Testing Turbulence Recover ability

In this chapter we talk about the assumptions.

3.1 Introduction

Add your text here.

Hello I am here. The following publication has been incorporated as Chapter 4.

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Chapter 4

Methodologies for Testing Turbulence Recover ability

In this chapter we talk about the assumptions.

4.1 Introduction

Add your text here.

Chapter 5

Conclusion

Conclude your thesis.

Appendix A

Appendix

Write your appendix here. Following two are examples.

A.1 Name of Appendix-1

A.2 Name of Appendix-2

Appendix B

Example of Citations

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Appendix C

Example of Code

C.1 Find the greatest number from a list of numbers in *Python*

```
a=[1,2,3,4,6,7,99,88,999]
max= 0
for i in a:
    if i > max:
        max=i
print(max)
```


Appendix D

Example of Equations

$$E = mc^2 \tag{D.1}$$

$$a = b + c \tag{D.2}$$

$$x = y + z \tag{D.3}$$

$$a = b + c \\ + d + e$$

$$\sin x + \cos x = 1 \tag{D.4}$$

Appendix E

Example of Figures



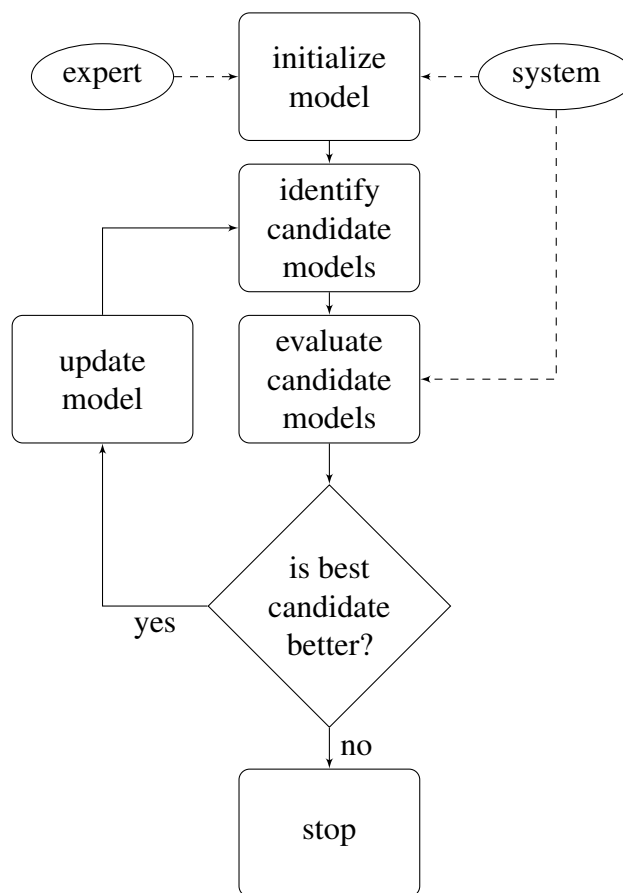
Figure E.1: The University Of Queensland

The Figure E.1 represents beauty of the UQ campus.

Appendix F

Example of Flow Charts

Figure F.1: Flow Chart



Flow chart F.1 is a simple example.

Appendix G

Example of Tables

Here is a really simple table G.1.

Table G.1: Name of the Australian Cities

Number	Name
1	Brisbane
2	Sydney
3	Melbourne
4	Canberra
5	Perth
6	Adelaide
7	Hobart
8	Darwin

Endquote goes here.

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